

**IMPLEMENTATION**

**Aim:**

To implement the given project based on Agile Methodology.

**Procedure:**

**Step 1: Set Up an Azure DevOps Project**

- Log in to Azure DevOps.
- Click "New Project" → Enter project name → Click "Create".
- Inside the project, navigate to "Repos" to store the code.

**Step 2: Add Your Web Application Code**

- Navigate to Repos → Click "Clone" to get the Git URL.
- Open Visual Studio Code / Terminal and run:  

```
git clone <repo_url>
cd <repo_folder>
```
- Add web application code (HTML, CSS, JavaScript, React, Angular, or backend like Node.js, .NET, Python, etc.).
- Commit & push:  

```
git add .
git commit -m "Initial commit"
git push origin main
```

**Step 3: Set Up Build Pipeline (CI/CD - Continuous Integration)**

- Navigate to Pipelines → Click "New Pipeline".
- Select Git Repository (Azure Repos, GitHub, or Bitbucket).
- Choose Starter Pipeline or a pre-configured template for your framework.
- Modify the azure-pipelines.yml file (Example for a Node.js app):

```
trigger:
- main

pool:
vmImage: 'ubuntu-latest'

steps:
- task: UseNode@1
  inputs:
    version: '16.x'

- script: npm install
  displayName: 'Install dependencies'

- script: npm run build
  displayName: 'Build application'

- task: PublishBuildArtifacts@1
  inputs:
    pathToPublish: 'dist'
    artifactName: 'drop'
```

Click "Save and Run" → The pipeline will start building app.

#### Step 4: Set Up Release Pipeline (CD - Continuous Deployment)

- Go to Releases → Click "New Release Pipeline".
- Select Azure App Service or Virtual Machines (VMs) for deployment.
- Add an artifact (from the build pipeline).
- Configure deployment stages (Dev, QA, Production).
- Click "Deploy" to push your web app to Azure.

#### **Result**

Thus the application was successfully implemented.